Network Theory

Unit-3 : Transformers

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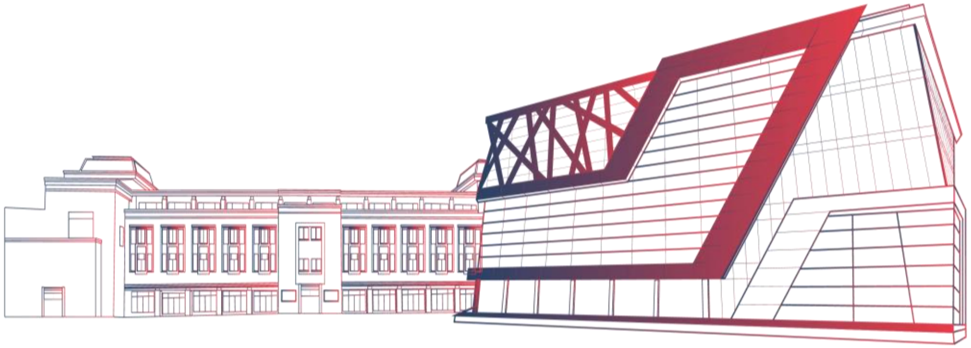


Multi-Disciplinary Department

Topics Covered

- Transformers
- Construction
- Working principle
- Emf equation,
- Transformer on No load and On load

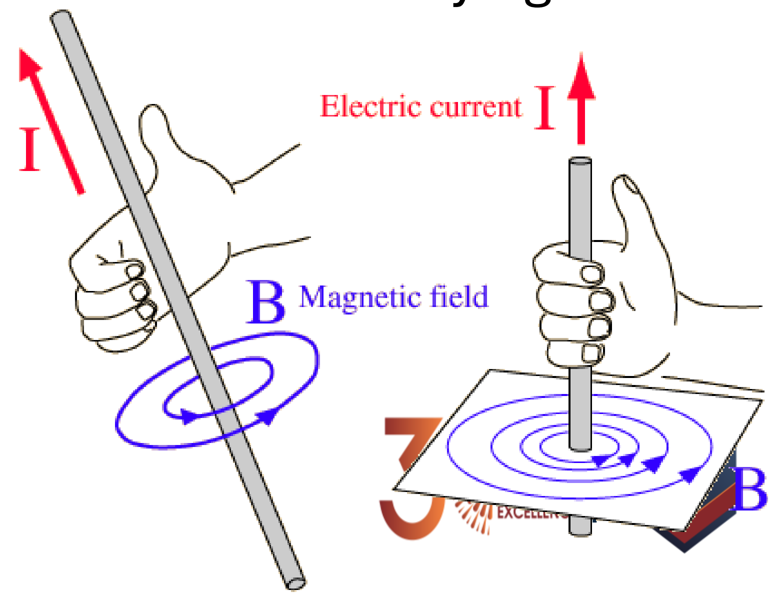
Magnetic Field around a Long Straight Current Carrying Wire (Ampere's Law)

- A current-carrying wire produces a magnetic field
- Right Hand Rule
  - Grasp the wire in your right hand
  - Point your thumb in the direction of the current
  - Your fingers will curl in the direction of the field
- The magnitude of the field at a distance r from a wire carrying a current of I is:

$$B = \frac{\mu_o I}{2\pi r}$$

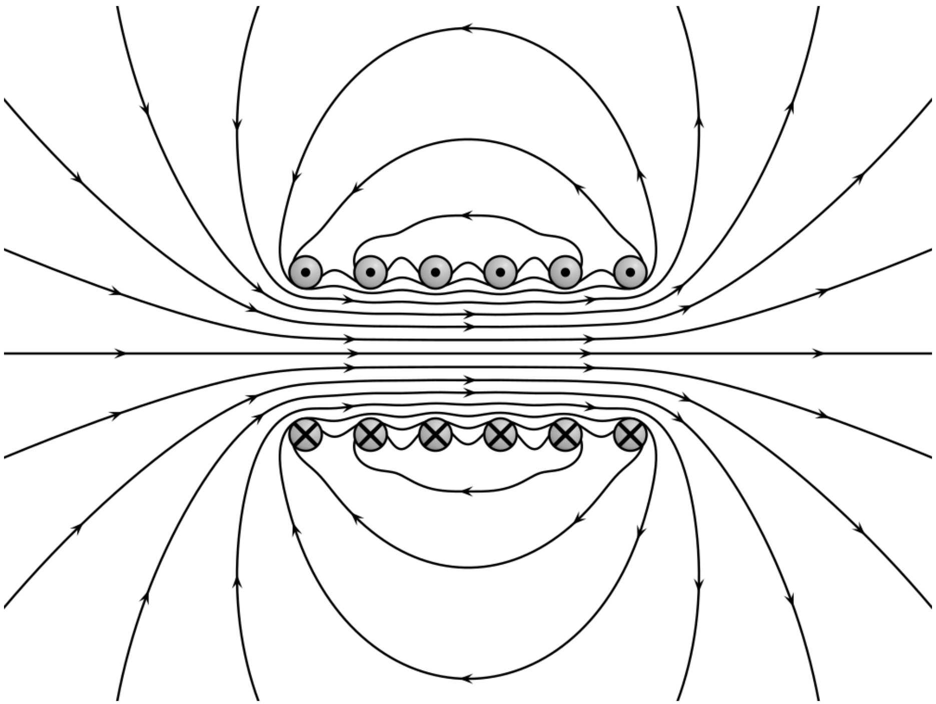
$\mu_o = 4\pi \times 10^{-7}$ T·m / A

μ_o is called the *permeability of free space*



Electromagnet Concept

- An electromagnet consists of an iron core placed inside a wire coil. The magnetic field strength of a wire coil carrying an electric current increases in direct proportion to the number of turns of the coil



Transformer

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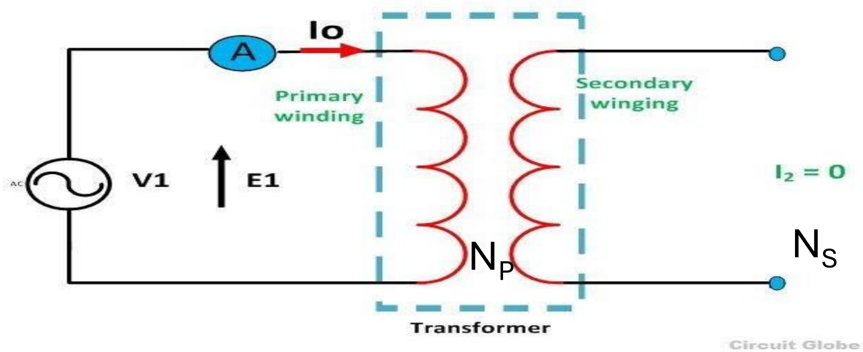
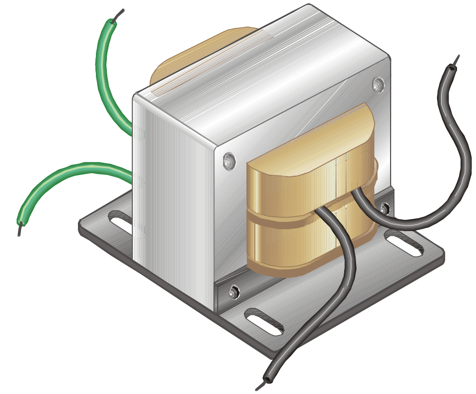
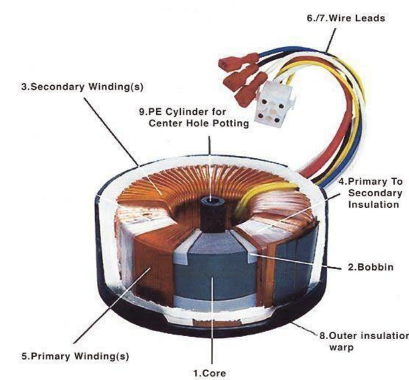
An A.C. device used to change high voltage low current A.C. into low voltage high current A.C. and vice-versa without changing the frequency







In brief,

1. Transfers electric power from one circuit to another
2. It does so without a change of frequency
3. It accomplishes this by electromagnetic induction
4. Turns ratio = N_S/N_P

   where: N_S = Number of turns in Secondary and

   N_P =Number of turns in Primary

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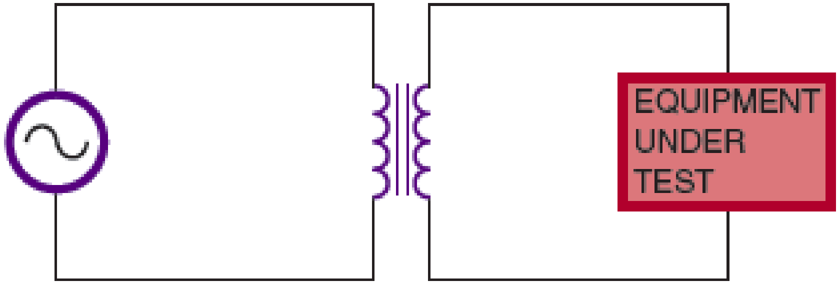
Transformer Uses

- AC power transmission
  - *STEP-UP Transformer*
  - Produces a secondary voltage greater than its primary voltage
  - Turns ratio is always greater than one
  - *STEP-DOWN Transformer*
  - Produces secondary voltage less than its primary voltage
  - Turns ratio is always less than one
- Impedance matching
- Electrical Isolation



Electromagnetic Induction Theory

Transformers behaviour is based on Faraday's Law of Induction

$$\mathcal{E} = -N \frac{d\Phi_B}{dt}$$

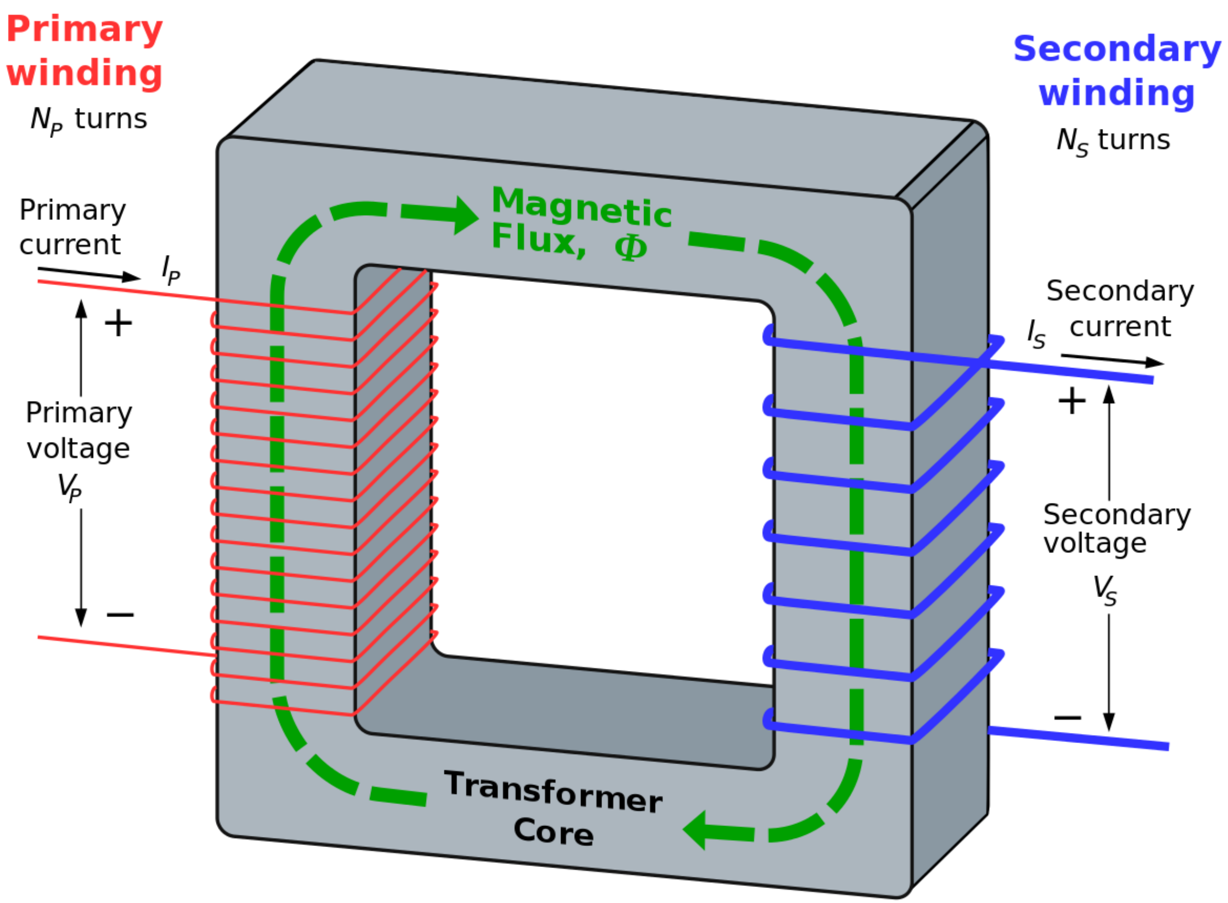
Where:-

ε – EMF (V)

N – No of turns of wire

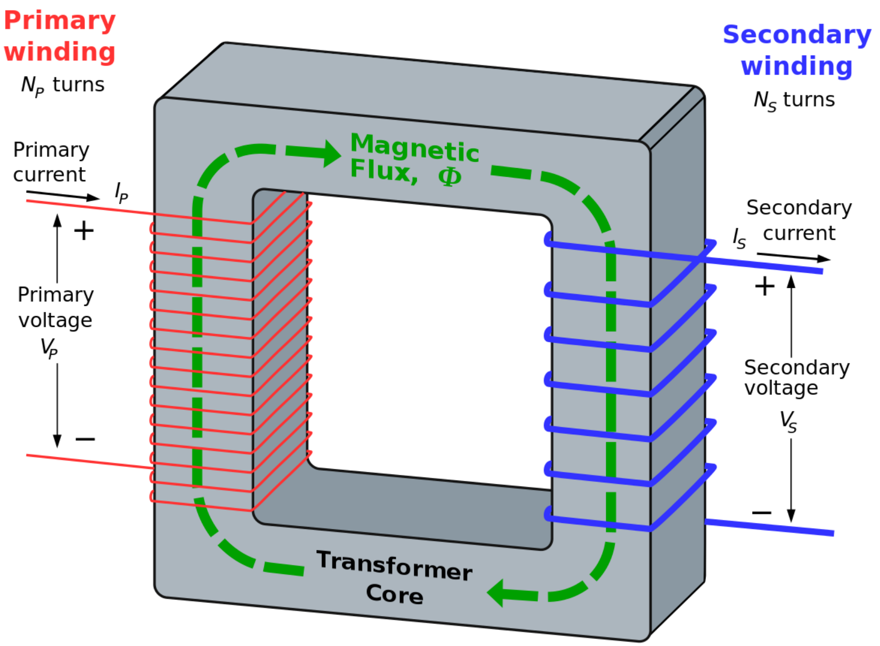
ΦB – Magnetic flux (Wb)

Induction Theory



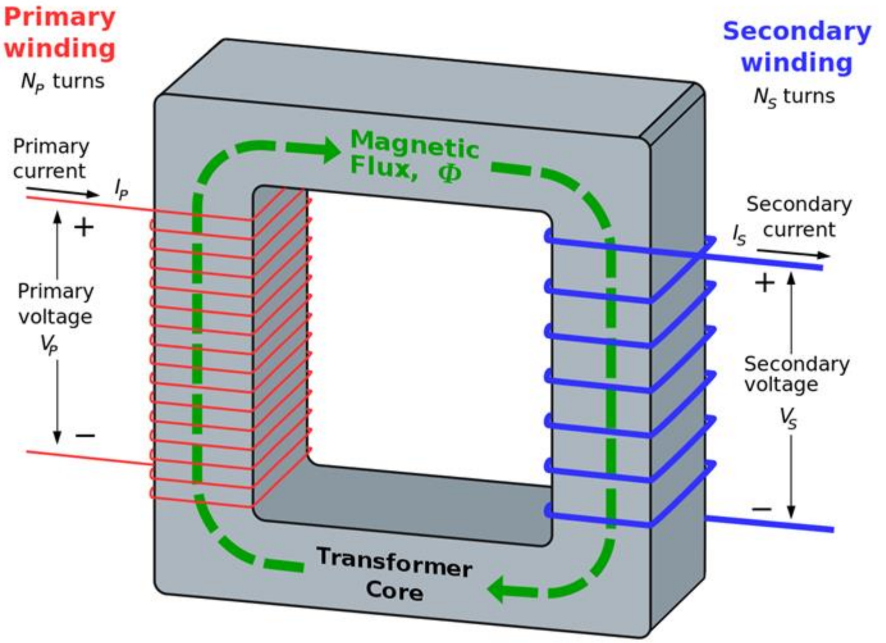
Working Principle of a Transformer

1. When current in the primary coil changes being alternating in nature, a changing magnetic field is produced
2. This changing magnetic field gets associated with the secondary through the soft iron core
3. Hence magnetic flux linked with the secondary coil changes.
4. Which induces e.m.f. in the secondary.



Basic Construction Of Transformer

- A transformer consists of two inductive windings and a laminated steel core.
- The coils are insulated from each other as well as from the steel core.

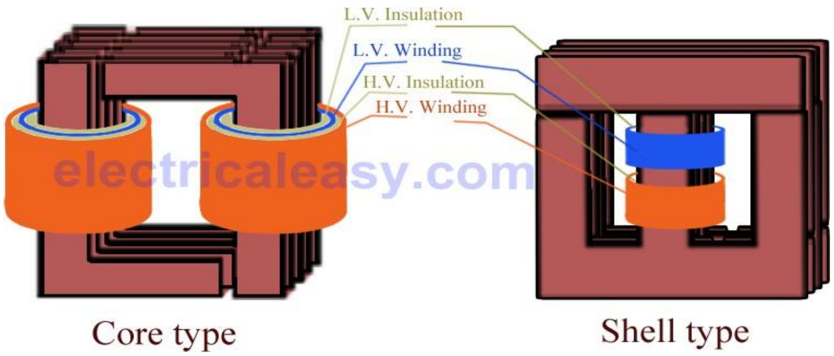


Types of Transformers

Transformers can be classified on different basis, like types of construction, types of cooling etc.

(A) On the basis of construction, transformers can be classified into two types as;

(i) Core type transformer and (ii) Shell type transformer, which are described below.



(i) Core type transformer

In core type transformer, windings are cylindrical former wound, mounted on the core limbs as shown in the figure above.

(ii) Shell type transformer

The coils are former wound and mounted in layers stacked with insulation between them.

Types of Transformers(Contd.)

(

(B)On the basis of their purpose

1.Step up transformer: Voltage increases (with subsequent decrease in current) at secondary.

2.Step down transformer: Voltage decreases (with subsequent increase in current) at secondary.

(C)On the basis of type of supply

1.Single phase transformer

2.Three phase transformer

(D)On the basis of their use

1.Power transformer: Used in transmission network, high rating

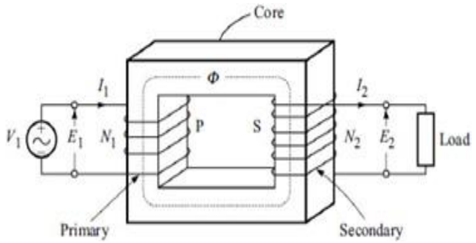
2.Distribution transformer: Used in distribution network, comparatively lower rating than that of power transformers.

1.Instrument transformer: Used in relay and protection purpose in different instruments in industries

- Current transformer (CT)
- Potential transformer (PT)
-

EMF Equation Of a Transformer

Consider a sinusoidally varying Voltage applied to the primary of the V_1 transformer shown in Fig .



.

Due to this voltage, a sinusoidally varying magnetic flux is set up in the core which can be represented as

$$\Phi = \Phi_m sin\, \omega t = \Phi_m sin\, 2\pi f t$$

Where Φ_m is the peak value of the flux and $\boldsymbol{f}$ is the frequency of sinusoidal variation of flux. As per the law of electromagnetic induction, the induced EMF in a winding of **N** turns is given as

$$e = -N\frac{d\Phi}{dt} = -N\frac{d}{dt}(\Phi_m\, sin\, sin\, \omega t\,)$$

$$= -N\omega\Phi_m\, cos\, cos\, \omega t$$

$$= \omega N\Phi_m sin(\omega t - \pi/2)$$

Thus, the peak of value of induced EMF is $E_m = \omega N\Phi_m$. Therefore the RMS value of induced EMF is

$$E = \frac{E_m}{\sqrt{2}} = \frac{\omega N\Phi_m}{\sqrt{2}} \quad = \frac{2\pi f N\Phi_m}{\sqrt{2}} = 4.44 f N\Phi_m$$

$$E = 4.44 f N\Phi_m$$

This equation is known as the equation of EMF of transformer, which can be used to find the EMF induced in any winding (primary or secondary winding) linking with flux Φ.

Continued...

Voltage ratios of the transformer with load:

- As shown in fig. let

N_1 = Number of turns in primary winding
N_2 = Number of turns in secondary winding

E_1 = rms induced voltage in primary winding

E_2 = rms induced voltage in secondary winding

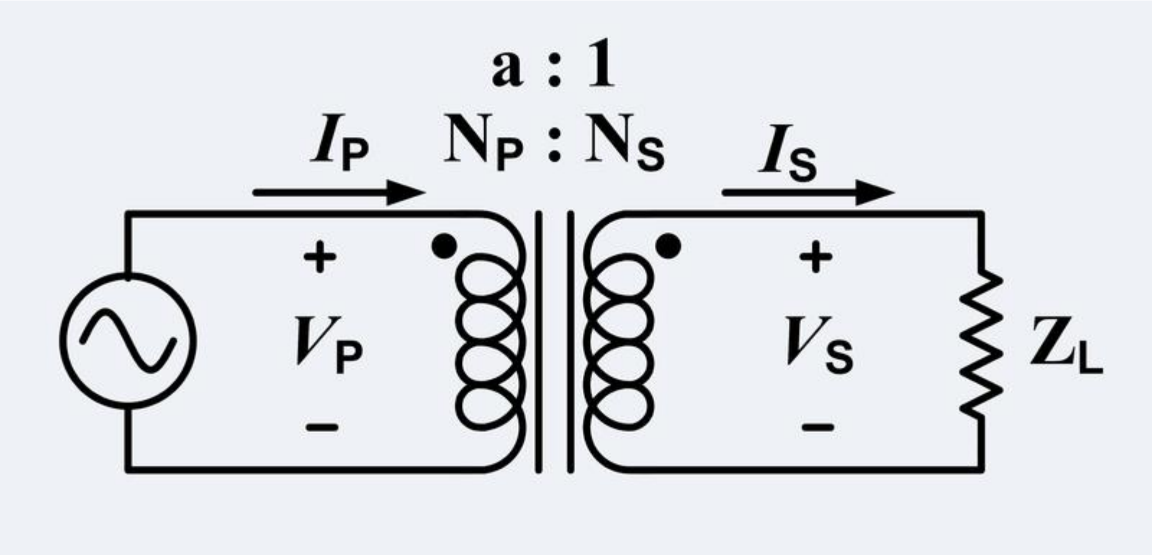
$E_1 = 4.44 f N_1 \Phi_m$ volts

$E_2 = 4.44 f N_2 \Phi_m$ volts

By taking the ratio of these expressions, we get,

$$\frac{E_1}{E_2} = \frac{N_1}{N_2}$$

Transformer Models



N_p = No of windings on the primary N_s = No of windings on the secondary
i_p = Current into the primary i_s = Current out from the secondary
V_p = Voltage across the primary V_s = Voltage across the secondary

Primary and Secondary Relationship

Turns ratio (a) is defined as the ratio of the number of turns in the secondary winding (N_{sec}) to the number of turns in the primary winding (N_{pri}) and Ip and Is are primary and secondary currents.

$$Vp/Vs = Np/Ns = Is/Ip = a$$

Note; $a < 1$ = Step up transformer

$a > 1$ = Step down transformer

Voltage and current angles are NOT affected hence,

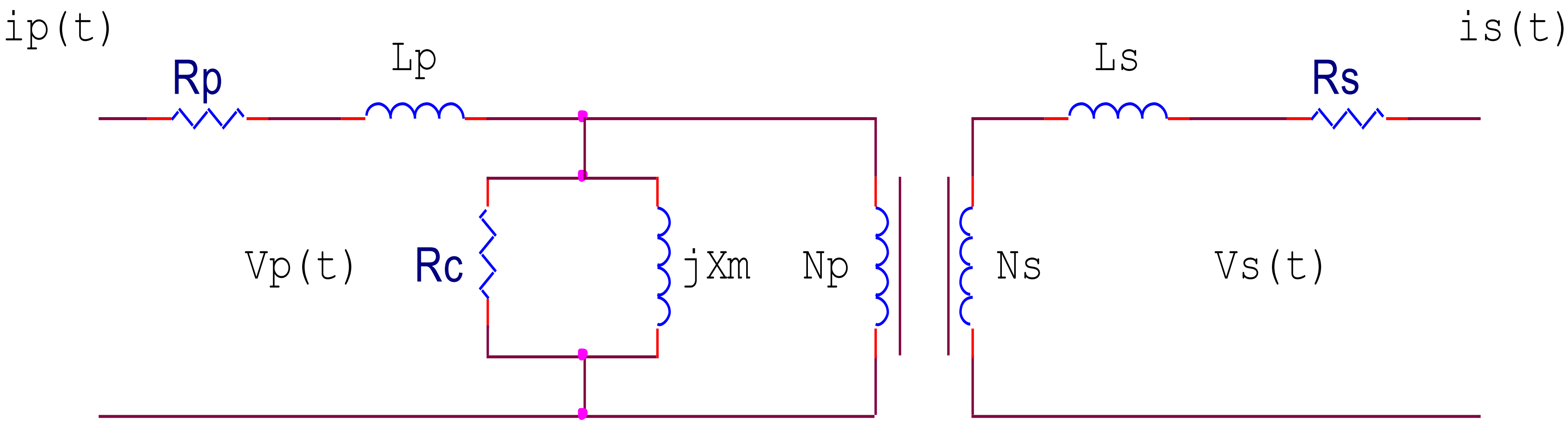
$\theta p = \theta s = \theta$

Losses in a Transformer

Transformers have losses and these losses must come into consideration as follows.

- Copper losses ($I^2 R$)
- Leakage Flux losses
- Core losses
    - Eddy currents
    - Hysteresis losses

Model for Transformer Losses



- **Copper losses (I^2 R)**
- **Leakage Flux losses**
- **Core losses**
  - **Eddy currents**
  - **Hysteresis losses**

Efficiency

- Efficiency (η) is the ratio of the power out to the power in of a transformer.
  - η in an *Ideal transformer*, no power losses
    - Pin = Vp Ip cos θ
    - Pout = Vs Is cos θ
    - Pin =Pout = Vp Ip cos θ = Vs Is cos θ
    - ηη-ideal = 100%
    - η in a non-ideal *transformer*, with iron and copper losses and power factor

The efficiency of a non –ideal transformer at any load is given by

%η = (x . full-load kVA . pf) / (x . full-load kVA .pf+ Wi + x^2 . Wcu) . 100
where x = ratio of actual to full load kVA
Wi = iron loss in kw
Wcu = full-load copper loss in kw

Voltage Regulation

Voltage regulation (VR) is the ability of a system to provide near constant voltage over a wide range of load conditions. Also it compares the VO at no load to VO at full load.

$$\boxed{VR = \frac{V_{nl} - V_{fl}}{V_{fl}} \times 100\%}$$

An Ideal transformer has a voltage regulation,

VR = 0%

Autotransformer

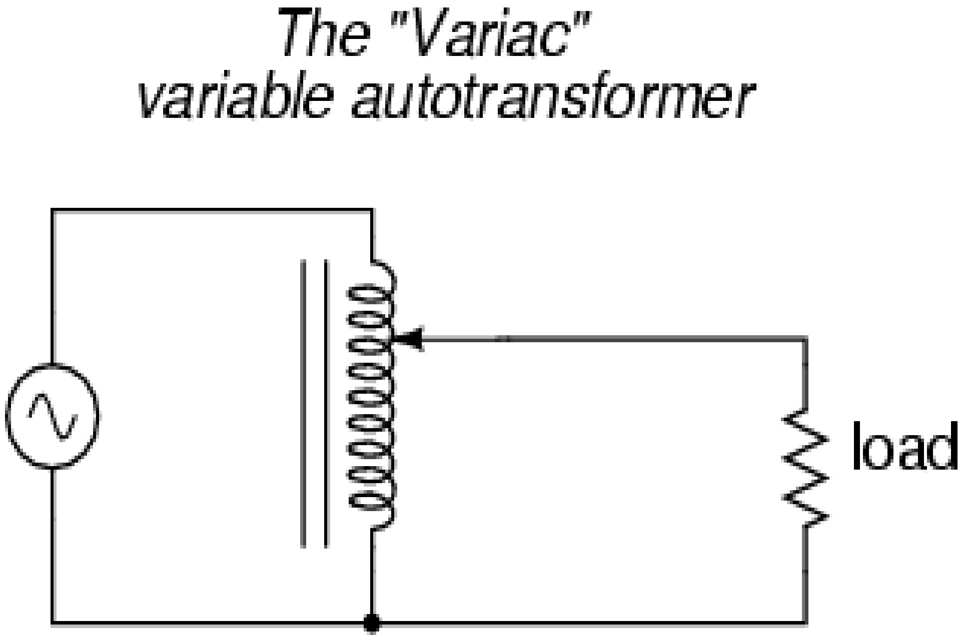
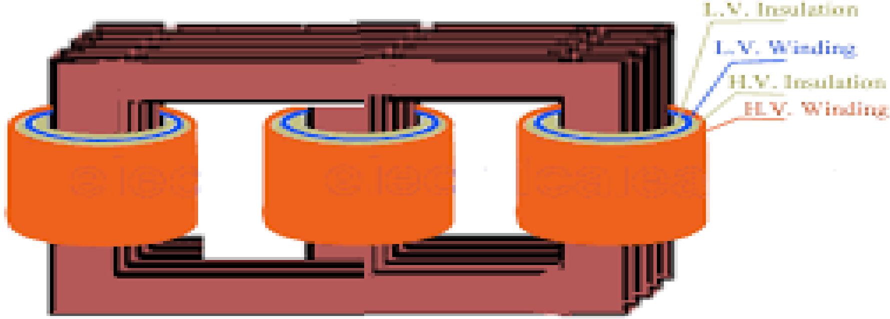
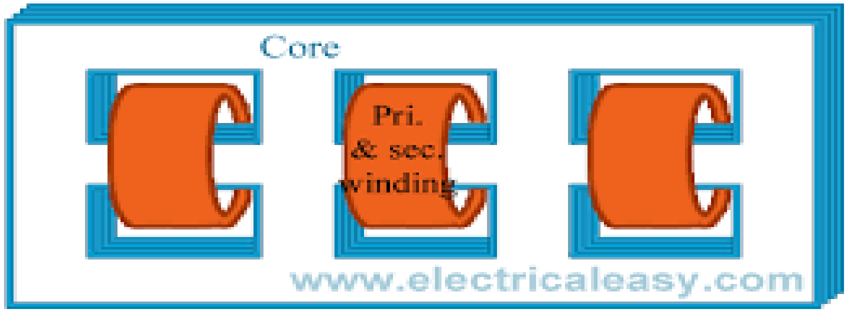


Fig. Autotransformer equivalent circuit

Three Phase Transformers



Core-type 3-phase Transformer



Shell type three phase transformer

Practice Questions

Q.1:What will be the secondary voltage at no load, if the primary of a 5 kVA, 220/110 V, 50 Hz transformer is fed at (i) 110 V, 50 Hz, and (ii) 220 V dc?

(*Ans.* V_2 = 55 V , V_2 = 0)

Q.2:It is desired to have 4.13 mWb maximum flux in the core of a transformer operating at 110 V and 50 Hz. Determine the required number of turns in the primary.

(*Ans.* N = 120)

Q.3:A 3000/200 V, 50 Hz, single-phase transformer has a cross-sectional area of 150 cm² for the core. If the number of turns on the low-voltage winding is 80, determine the number of turns on the high-voltage winding and maximum value of flux density in the core.

(*Ans.* Bm = 0.75 wb/m^2)

Q.4: A transformer has a turn ratio N_1:N_2 of 4. If a 50 ohm resistance is connected across the secondary, what is the resistance referred to primary?

(Ans. 800 Ohms)

Q.5: A 200 kVA, 2200/440 V, 50 Hz, single-phase transformer is operating at full load, 0.8 lagging pf. The voltage on secondary of the transformer at full load, 0.8 lagging pf is 400 V. Calculate voltage regulation of the transformer.

(Ans. 9.09%)

Happy Learning